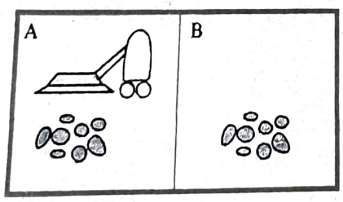
Year 2024-2025 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Date	17-12-2024	Maximum Marks	10+50
Course Code	IS3531A	Duration	2 hours
Sem	V	CIE-1(Retest)	
UG/PG	UG	Faculty: AS/VH/VG/JS/SHRS/ARA/MEM	

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Q. No.	QUIZ	M	BT	CO
1	An agent's _____ Is the complete history of everything the agent has ever perceived. percept sequence	1	2	1
2	What is rational at any given time depends on four factors. Which are they?	2	2	2
3	What is the Utility function of an agent ?	1	1	3
4	The _____implements the agent function. Agent program	1	1	1
5	A classification model serves two important roles in data mining. Which are they ? Predictive & Descriptive modelling	1	2	5
6	Suggest a method to avoid data fragmentation problem Pruning, min support thresholds, smoothing techniques	2	3	4
7	Comment on the effect of the choice of impurity measure on performance of decision tree classifiers → affects how decision tree split nodes. → entropy: leads to more balanced tree but comp expensive → gini index: simpler & faster but can lead to biased splits.	1	3	2
8	Give a method for reducing the effect of overfitting ? Early stopping cross-validation.	1	3	1

Q. No.	TEST	M	BT	CO
1 a	Define Artificial Intelligence from the perspective of (i) Acting Humanly and (ii) Acting Rationally. Consider a vacuum-cleaner world with just two locations as given in the figure below. Construct a partial tabulation of a simple agent function for the vacuum-cleaner world and explain the scenario.  S1 S2	5	2	1

A fn that assigns a numerical value to each possible states, indicating how desirable that state is for agent. It helps agent to make decisions that maximise expected utility.

* Performance measure that defines the success
* agent's prior knowledge about the env
* the actions available to the agent.
* the percept sequence.

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Nominal & Ordinal → classification models
Interval & Ratio → Regression models

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1 b	Using relevant examples , explain the different attribute types	5	1	1
2a	With the help of a neat diagram and pseudocode, explain a model based reflex agent	5	1	2
2 b	Explain any 5 characteristics of a decision tree	5	2	1
3 a	Characterise the following scenarios based on the standard features of the task environment (i) Shopping for used AI books on the Internet. (ii) Playing a tennis match. (iii) Practicing tennis against a wall. (iv) Autonomous Mars rover	10	3	2
4	Consider the following statements . State whether you agree with the statements or not with proper justification. (v) An agent that senses only partial information about the state cannot be perfectly rational (vi) Suppose an agent selects its action uniformly at random from the set of possible actions. There exists a deterministic task environment in which this agent is rational. (vii) There exists a task environment in which every agent is rational. (viii) The input to an agent program is the same as the input to the agent function. (ix) A perfectly rational poker-playing agent never loses.	10	4	4
5	Consider the below given data set Calculate : (i) The entropy of the above collection (ii) What are the information gains of a1and a2 relative to these training examples? (iii) What is the best split (between a1and a2) according to the Gini index?	10 2+4 +4	3	5

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−